A number of measures are used to assess forecasting accuracy. These include the traditional statistical accuracy measures: mean absolute error, root mean square error, mean absolute percentage error and Theil's inequality coefficient (Theil-U). Mean absolute percentage error and Theil-U enable evaluation of instances where forecast errors must be modeled independently of the variables. As such are they constructed to lie within the range [0,1], with zero indicating a perfect fit. The measure correct directional change assesses the ability of the strategy to predict the actual next period change of a forecast variable, an important issue in a trading strategy. Statistical performance measures used to analyze forecasting accuracy are detailed in Table 13.3. Please refer to Hanke and Reitsch (1998) and Pindyck and Rubinfeld (1998) for detailed discussion on these measures.

Forecasting accuracy measures enable us to understand the statistical underpinnings of our strategy. However, they not lend well into analysis

Table 13.3 Statistical Performance Measures

Performance Measure	Description
Mean Absolute Error	$MAE = \frac{1}{T}\sum_{t=1}^{T} \lvert\tilde{y}_t - y_t\rvert$ (13.32)
Mean Absolute Percentage Error	$MAPE = \frac{100}{T}\sum_{t=1}^{T} \left\lvert\frac{\tilde{y}_t - y_t}{y_t}\right\rvert$ (13.33)
Root Mean Square Error	$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T} (\tilde{y}_t - y_t)^2}$ (13.34)
Theil's Inequality Coefficient	$U = \dfrac{\sqrt{\frac{1}{T}\sum_{t=1}^{T} (\tilde{y}_t - y_t)^2}}{\sqrt{\frac{1}{T}\sum_{t=1}^{T} (\tilde{y}_t)^2} + \sqrt{\frac{1}{T}\sum_{t=1}^{T} (y_t)^2}}$ (13.35)
Correct Directional Change	$CDC = \frac{100}{N}\sum_{t=1}^{N} D_t$ (13.36) where $D_t = 1$ if $y_t \cdot \tilde{y}_t > 0$ else $D_t = 0$

Source: Dunis et al. (2004)
y_t is the actual change at time t.
$\tilde{y}_t$ is the forecast change.
t = 1 to t = T for the forecast period.